Do AI coding assistants make project ideas more alike?

A monthly study of new GitHub repositories

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AI could narrow or broaden project ideas

More alike

Shared models
Similar suggestions
Similar projects

More varied

Easier implementation
More people building
More ideas attempted

AI-assisted stories became more similar. We ask whether software does too. [3]

Compare monthly diversity before and after release

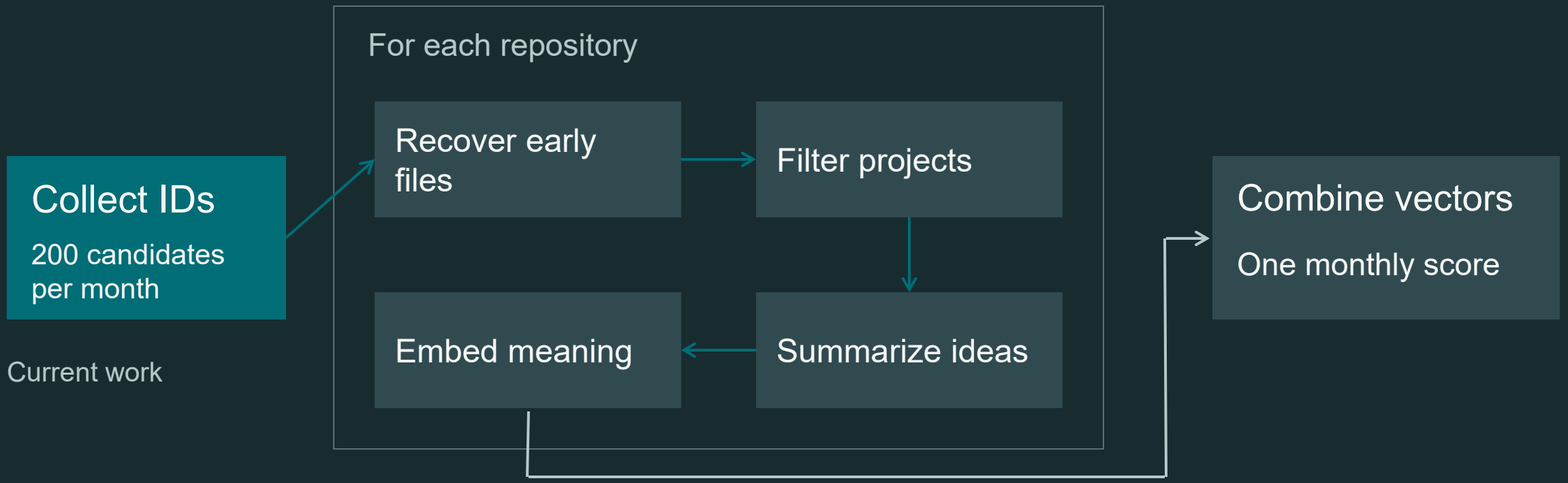


Planned: equal eligible samples

100–200 per month; set after pilot

ChatGPT: November 30, 2022. Copilot was already available. [2, 8]

Each month, turn repositories into one diversity score



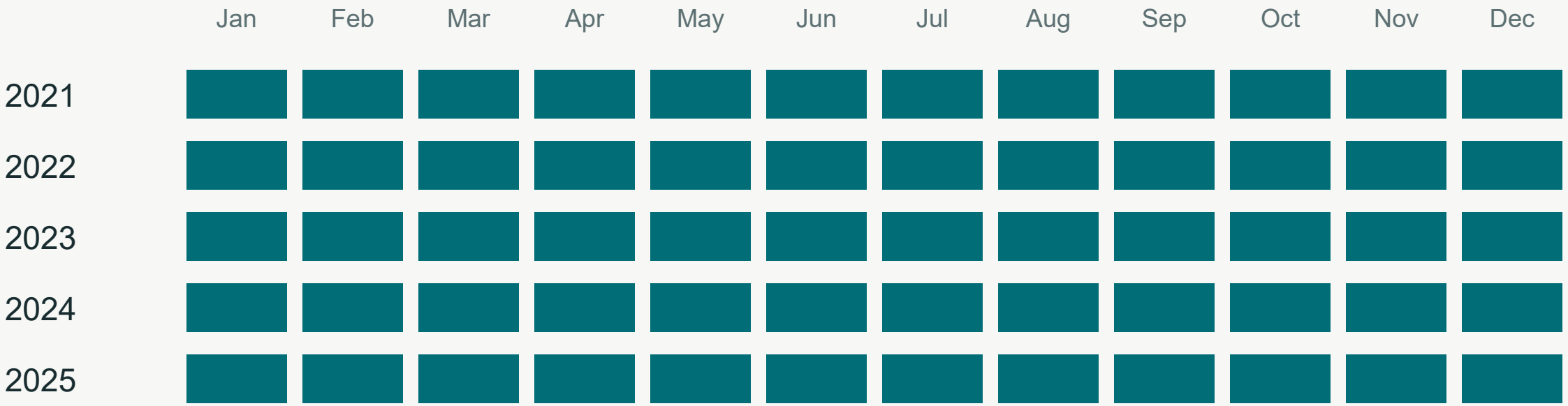
Current work

60 monthly scores → Trend test → Robustness checks

200 unfiltered candidates per month

60 months covered

12,000 repository IDs



IDs collected; eligibility has not been assessed.

The final eligible sample may require collecting more candidates.

Next: recover each project's early files

Creation events
Distinct repository IDs

→

Seeded ID order
Within each sampled month

→

Historical snapshot
Latest recoverable by day 30

Keep records

Repository ID · commit ID
Decision · evidence · cost

If unavailable

Log the failure.
Take the next candidate.

GH Archive locates creation events; the GitHub API supplies repository contents. [4, 5]

Next: filter for documented software projects

Rule pass

Remove forks, mirrors and profiles
Require source code + English README
Reject short or duplicate READMEs

GLM 5.3 pass

Keep “project” only
Exclude tutorials, coursework, templates, non-software and unclear cases

Require a supporting quote. Record each decision and reason.

README minimum: 30 words. Fixed prompt; identity and date redaction.

Describe each project before comparing ideas

Early files

README + source evidence

→

Idea statement

Purpose · users · function

→

Embedding

Similar meanings sit close

Up to 80 words

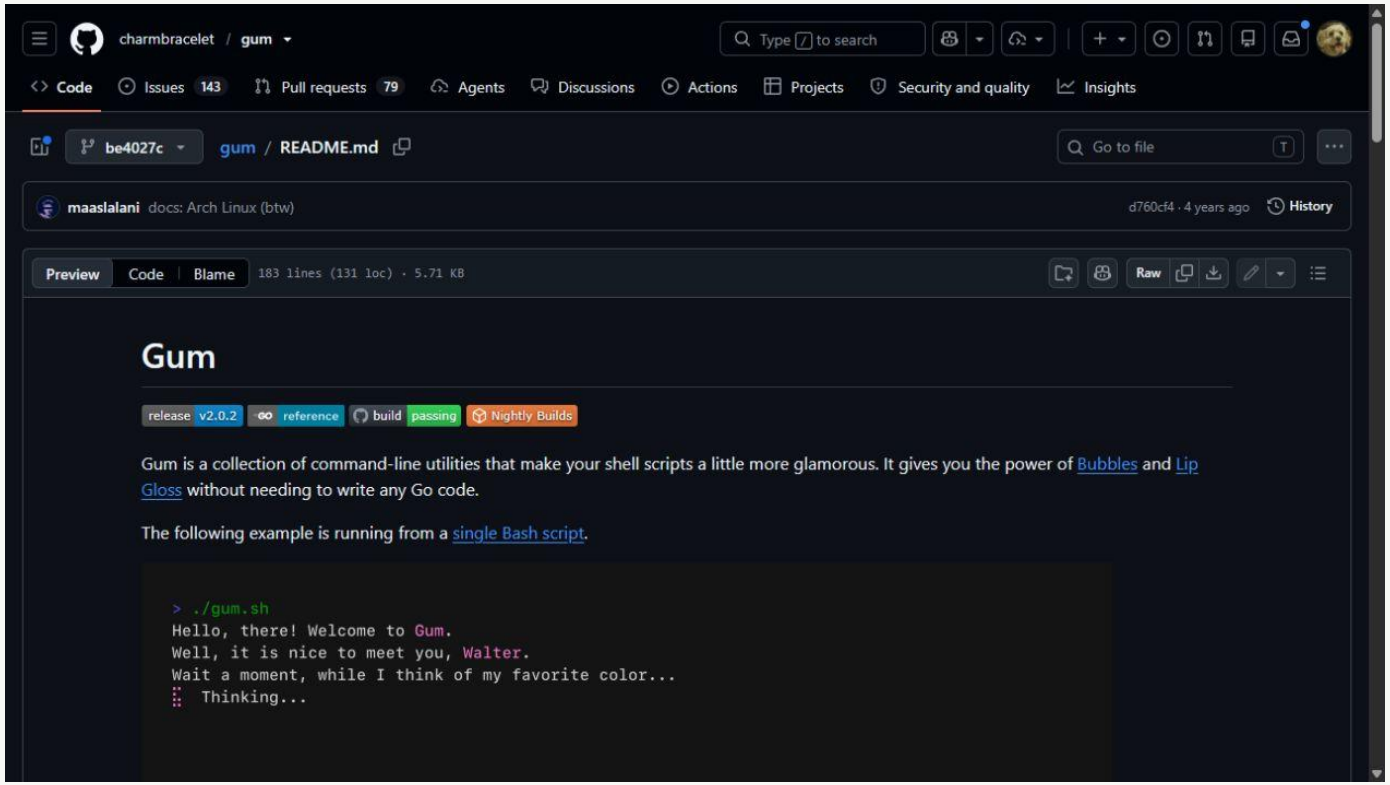
Supporting passages required

Unclear purpose: insufficient evidence

Planned: fixed summary prompt; all-MiniLM-L6-v2 embeddings. [9, 10]

Turn repository details into an idea statement

charmbracelet/gum



Illustrative LLM summary

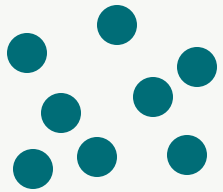
Gum helps shell-script authors add interactive prompts and styled terminal output without writing Go.

It provides commands for text input, fuzzy selection, progress indicators, and text layout.

Early commit be4027c · demonstration only; summary extraction is planned. [11]

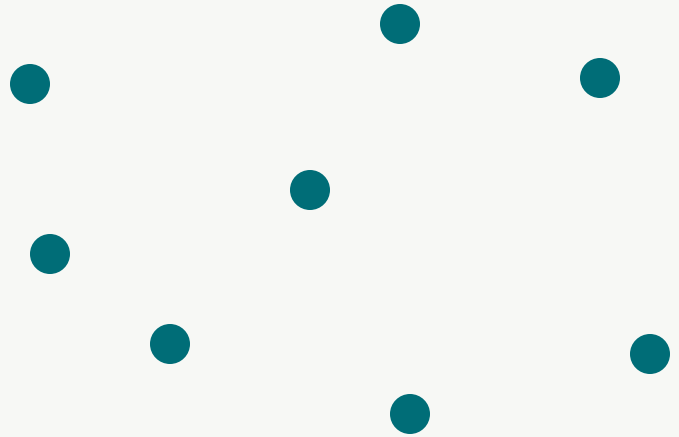
More varied ideas produce a higher monthly score

Similar ideas



Lower average distance

Varied ideas

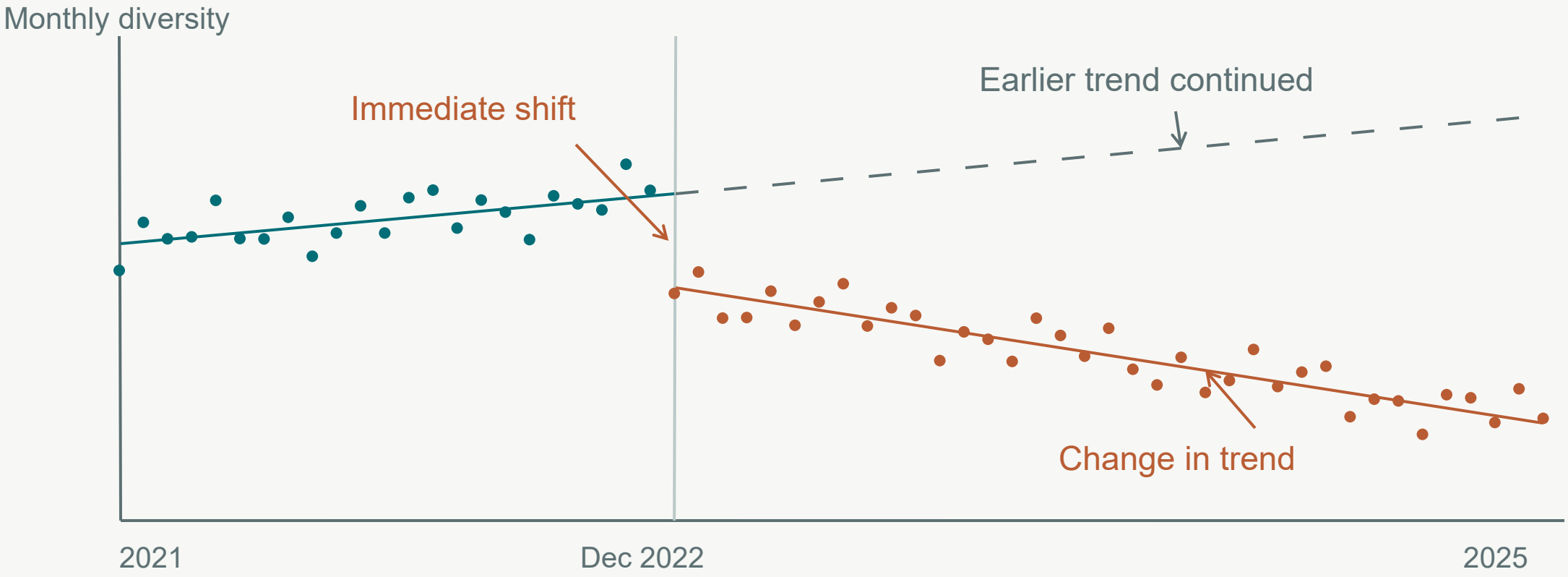


Higher average distance

Idea vectors → Compare every pair → Average within the month

Schematic of similarity; not study data or a 2D analysis.

Test whether the diversity trend changes after release



Estimate the shift and trend change, each with a 95% confidence interval.

Illustration only. Annual patterns are accounted for in the model.

Validate the filter and the diversity measure

100	candidates	Is “project” classified correctly? Precision + recall, before vs. after
50	summaries	Does the statement match its source? Target: 90% faithful summaries
50	idea pairs	Does distance match human judgment? 1–5 ratings; agreement + correlation

Planned: two independent reviewers; revise failed measurements before the full run.

Check robustness without claiming causation

Rerun the same model

Exclude AI-topic projects
Keep one repository per owner
Embed cleaned READMEs

The question we can answer

Whether diversity changed around the release

AI use is unobserved.
Other changes may explain the pattern.

Scope: accessible, documented, English-language projects. Track missing snapshots by month.

The scraping pipeline is the current milestone

Deliverable	Current status / next target	Lead
Data pipeline	Scraping setup; 200 IDs per month	Samuel
Representation + analysis	Next: summaries, embeddings, scores	Spencer
Summary validation	Planned: source checks	Hezeki
Similarity validation	Planned: human pair ratings	Hezeki

Next: retrieve early files, filter candidates, validate decisions

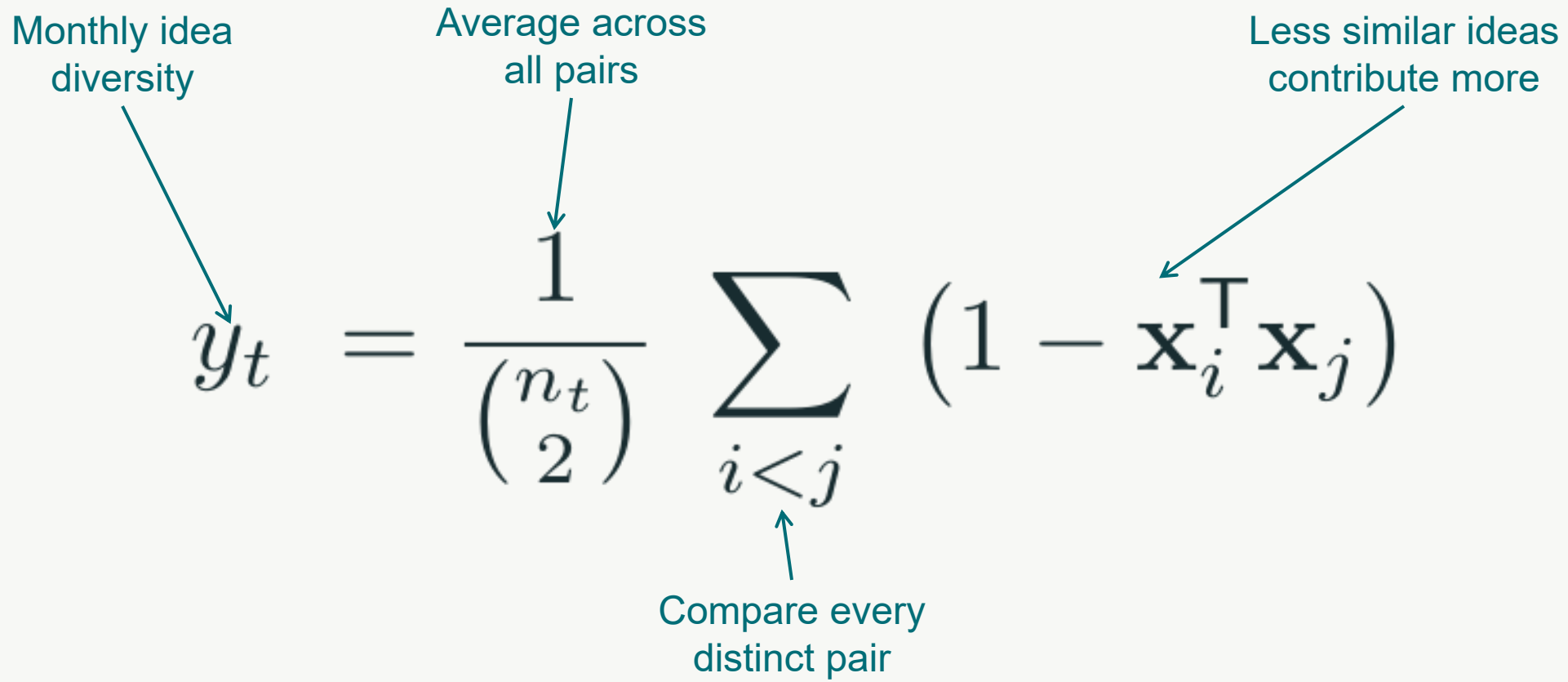
All three members contribute; leads coordinate each component.

Validate measurement, complete the sample, then test

By Oct 27	Set up scraping; collect IDs	Samuel
Oct 28–Nov 8	Validate filtering; freeze prompts	Spencer
Nov 9–20	Complete samples + human reviews	Spencer
Nov 21–final	Fit trend; rerun; write + present	Hezeki

A reliable answer can be lower, higher, or no clear change.

Average idea differences into one monthly score



Each x is a unit-length idea vector; n is the number of projects in that month.

Fit the full model to the 60 monthly scores

Trend before release

Release-date shift

Change in monthly trend

$$y_t = \beta_0 + \beta_1 t + \beta_2 \,\text{post}_t + \beta_3 (t - 24)\,\text{post}_t$$

$$+ \gamma_1 \sin\left(\frac{2\pi t}{12}\right) + \gamma_2 \cos\left(\frac{2\pi t}{12}\right) + \varepsilon_t$$

Recurring yearly patterns

Remaining variation

t = 24 is December 2022; post switches from 0 to 1 at that month.

References · context and data

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[3] Anil R. Doshi and Oliver P. Hauser. 2024. Generative AI enhances individual creativity but reduces the collective diversity of novel content. Science Advances 10, 28, eadn5290. doi:10.1126/sciadv.adn5290

[4] GH Archive. n.d. GH Archive. https://www.gharchive.org/

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References · measurement and inference

[6] James Lopez Bernal, Steven Cummins, and Antonio Gasparrini. 2017. Interrupted time series regression for the evaluation of public health interventions: a tutorial. Int. J. Epidemiology 46, 1, 348–355. doi:10.1093/ije/dyw098

[7] Whitney K. Newey and Kenneth D. West. 1987. A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. Econometrica 55, 3, 703–708. doi:10.2307/1913610

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[10] Sentence Transformers. n.d. all-MiniLM-L6-v2. Hugging Face model card. https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

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